



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR Symbiosis International (Deemed University)

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Practical No.: 7

Aim –

Perform DOM traversal and updates; develop a To-Do List application with add, edit, and delete features. Implement a registration form using JavaScript event handling and form validation.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

```
<!DOCTYPE html>
<html>
<head>
  <title>To Do List</title>

  <style>
    body {
      font-family: Arial;
      background: #f2f2f2;
      text-align: center;
    }

    .container {
      width: 400px;
      margin: 50px auto;
      background: white;
      padding: 20px;
      border-radius: 8px;
    }

    input {
      padding: 10px;
```

```

        width: 220px;
    }

    button {
        padding: 10px;
        margin: 5px;
        border: none;
        cursor: pointer;
    }

    #addBtn {
        background: green;
        color: white;
    }

    .edit {
        background: orange;
    }

    .delete {
        background: red;
        color: white;
    }

    li {
        list-style: none;
        margin: 10px 0;
        padding: 10px;
        background: #eee;
        display: flex;
        justify-content: space-between;
        align-items: center;
    }
</style>
</head>

<body>

<div class="container">

    <h1>To Do List</h1>

    <input type="text" id="taskInput" placeholder="Enter task">
    <button id="addBtn">Add</button>

    <ul id="taskList"></ul>

```

</div>

<script>

```
let input = document.getElementById("taskInput");
let addBtn = document.getElementById("addBtn");
let taskList = document.getElementById("taskList");

addBtn.addEventListener("click", function() {

    let task = input.value;

    if (task === "") {
        alert("Enter a task");
        return;
    }

    let li = document.createElement("li");
    let span = document.createElement("span");
    let editBtn = document.createElement("button");
    let deleteBtn = document.createElement("button");

    span.innerText = task;

    editBtn.innerText = "Edit";
    editBtn.className = "edit";

    deleteBtn.innerText = "Delete";
    deleteBtn.className = "delete";

    editBtn.addEventListener("click", function() {

        let newTask = prompt("Edit task:", span.innerText);

        if (newTask !== null && newTask !== "") {
            span.innerText = newTask;
        }

    });

    deleteBtn.addEventListener("click", function() {

        li.remove();

    });

    li.appendChild(span);
    li.appendChild(editBtn);
    li.appendChild(deleteBtn);
```

```
// Add li to ul
taskList.appendChild(li);

// Clear input
input.value = "";
});

</script>

</body>
</html>
```

Output –

The screenshot displays a web application titled "To Do List". At the top, there is a text input field with the placeholder text "Enter task" and a green "Add" button. Below this, there is a list of four tasks, each displayed in a light gray box. Each task box contains the task name, an orange "Edit" button, and a red "Delete" button. The tasks listed are "cooking", "Watering Plants", "Assignment of Python", and "sejal lende 24070521035".

Task	Edit	Delete
cooking	Edit	Delete
Watering Plants	Edit	Delete
Assignment of Python	Edit	Delete
sejal lende 24070521035	Edit	Delete

Case Study Title –

Implement a registration form using JavaScript event handling and form validation.

Case Study Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Create Account</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      min-height: 100vh;
      font-family: "Segoe UI", system-ui, -apple-system, sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      padding: 40px 20px;
      background: linear-gradient(135deg, #f0fdfa 0%, #ecfeff 50%, #f0f9ff 100%);
    }

    .form-container {
      width: 100%;
      max-width: 860px;
      background: #ffffff;
      border-radius: 20px;
      padding: 40px 50px;
      box-shadow: 0 20px 50px rgba(13, 148, 136, 0.12);
    }

    .form-header {
      text-align: center;
      margin-bottom: 32px;
    }

    .form-header h2 {
      font-size: 28px;
      color: #0f766e;
      font-weight: 700;
      margin-bottom: 6px;
    }

    .form-header p {
      font-size: 14px;
      color: #64748b;
    }

    .section-title {
      font-size: 12px;
      font-weight: 700;
      text-transform: uppercase;
      letter-spacing: 1.1px;
      color: #0d9488;
      margin: 24px 0 14px;
```

```

    display: flex;
    align-items: center;
    gap: 12px;
}

.section-title::after {
    content: "";
    flex: 1;
    height: 1px;
    background: #ccfbf1;
}

.form-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 16px 20px;
}

.full {
    grid-column: 1 / -1;
}

label {
    display: block;
    font-size: 13px;
    font-weight: 600;
    color: #334155;
    margin-bottom: 6px;
}

.required {
    color: #e11d48;
}

.input-box {
    position: relative;
}

.input-icon {
    position: absolute;
    left: 13px;
    top: 50%;
    transform: translateY(-50%);
    color: #94a3b8;
    font-size: 13px;
    pointer-events: none;
}

input, select {
    width: 100%;
    height: 44px;
    border: 1.5px solid #e2e8f0;
    border-radius: 10px;
    padding: 0 14px 0 36px;
    background: #f8fafc;
    font-size: 13.5px;
    color: #1e293b;
    transition: 0.2s;
}

select {
    padding-left: 12px;
    cursor: pointer;
}

input:focus, select:focus {
    outline: none;
    border-color: #14b8a6;
    background: #ffffff;
    box-shadow: 0 0 0 3px rgba(20, 184, 166, 0.15);
}

.birthday {

```

```

    display: grid;
    grid-template-columns: 1fr 1.4fr 1fr;
    gap: 12px;
}

.input-error {
    border-color: #e11d48 !important;
    background: #fef2f2 !important;
}

.input-success {
    border-color: #10b981 !important;
}

.error-text {
    min-height: 15px;
    margin-top: 4px;
    font-size: 11.5px;
    color: #e11d48;
    font-weight: 500;
}

.terms {
    display: flex;
    align-items: center;
    gap: 10px;
    margin-top: 22px;
    padding: 13px 16px;
    background: #f0fdfa;
    border: 1px solid #99f6e4;
    border-radius: 10px;
}

.terms input {
    width: 16px;
    height: 16px;
    accent-color: #0d9488;
    cursor: pointer;
}

.terms label {
    margin: 0;
    font-size: 13px;
    color: #475569;
    cursor: pointer;
}

.terms-error {
    border-color: #e11d48 !important;
    background: #fef2f2 !important;
}

button {
    width: 100%;
    height: 48px;
    margin-top: 20px;
    border: none;
    border-radius: 11px;
    background: linear-gradient(135deg, #0d9488, #0f766e);
    color: white;
    font-size: 15px;
    font-weight: 600;
    cursor: pointer;
    transition: 0.2s;
    box-shadow: 0 8px 20px rgba(13, 148, 136, 0.25);
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 12px 28px rgba(13, 148, 136, 0.3);
}

.designer {

```

```

        text-align: center;
        margin-top: 22px;
        font-size: 12.5px;
        color: #64748b;
    }

    .designer strong {
        color: #0d9488;
        font-weight: 600;
    }

    /* Responsive */
    @media (max-width: 700px) {
        .form-container {
            padding: 30px 22px;
        }
        .form-grid {
            grid-template-columns: 1 fr;
        }
        .full {
            grid-column: auto;
        }
        .birthday {
            grid-template-columns: 1 fr;
        }
    }
}
</style>
</head>
<body>
    <div class="form-container">
        <div class="form-header">
            <h2>Create Account</h2>
            <p>Fill your information below</p>
        </div>

        <form id="registrationForm">
            <div class="section-title">Personal Information</div>

            <div class="form-grid">
                <div class="form-group">
                    <label>First Name <span class="required">*</span></label>
                    <div class="input-box">
                        <span class="input-icon">●</span>
                        <input type="text" id="firstname" placeholder="First name">
                    </div>
                    <p class="error-text" id="firstnameError"></p>
                </div>

                <div class="form-group">
                    <label>Last Name <span class="required">*</span></label>
                    <div class="input-box">
                        <span class="input-icon">●</span>
                        <input type="text" id="lastname" placeholder="Last name">
                    </div>
                    <p class="error-text" id="lastnameError"></p>
                </div>

                <div class="form-group full">
                    <label>Birthday <span class="required">*</span></label>
                    <div class="birthday">
                        <select id="day"><option value="">Day</option></select>
                        <select id="month"><option value="">Month</option></select>
                        <select id="year"><option value="">Year</option></select>
                    </div>
                    <p class="error-text" id="birthdayError"></p>
                </div>
            </div>

            <div class="section-title">Account Details</div>

            <div class="form-grid">
                <div class="form-group">
                    <label>Username <span class="required">*</span></label>

```

```

        <div class="input-box">
            <span class="input-icon">@</span>
            <input type="text" id="username" placeholder="Username">
        </div>
        <p class="error-text" id="usernameError"></p>
    </div>

    <div class="form-group">
        <label>E-mail <span class="required">*</span></label>
        <div class="input-box">
            <span class="input-icon">✉</span>
            <input type="text" id="email" placeholder="name@email.com">
        </div>
        <p class="error-text" id="emailError"></p>
    </div>

    <div class="form-group full">
        <label>Website <span class="required">*</span></label>
        <div class="input-box">
            <span class="input-icon">🌐</span>
            <input type="text" id="website" placeholder="https://example.com">
        </div>
        <p class="error-text" id="websiteError"></p>
    </div>

    <div class="form-group">
        <label>Password <span class="required">*</span></label>
        <div class="input-box">
            <span class="input-icon">●</span>
            <input type="password" id="password" placeholder="Password">
        </div>
        <p class="error-text" id="passwordError"></p>
    </div>

    <div class="form-group">
        <label>Re-password <span class="required">*</span></label>
        <div class="input-box">
            <span class="input-icon">●</span>
            <input type="password" id="repassword" placeholder="Confirm password">
        </div>
        <p class="error-text" id="repasswordError"></p>
    </div>

    <div class="terms" id="termsBox">
        <input type="checkbox" id="terms">
        <label for="terms">I agree to the terms & conditions</label>
    </div>
    <p class="error-text" id="termsError"></p>

    <button type="submit">Create My Account ♦</button>
</form>

<div class="designer">
    Made by <strong>Sejal Lende</strong> • 24070521035
</div>
</div>

<script>
    const form = document.getElementById("registrationForm");
    const firstname = document.getElementById("firstname");
    const lastname = document.getElementById("lastname");
    const username = document.getElementById("username");
    const email = document.getElementById("email");
    const website = document.getElementById("website");
    const password = document.getElementById("password");
    const repassword = document.getElementById("repassword");
    const terms = document.getElementById("terms");
    const termsBox = document.getElementById("termsBox");
    const day = document.getElementById("day");
    const month = document.getElementById("month");
    const year = document.getElementById("year");

```

```

for (let i = 1; i <= 31; i++) {
  let option = document.createElement("option");
  option.value = i;
  option.textContent = i;
  day.appendChild(option);
}

const months =
["January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"];
months.forEach(m => {
  let option = document.createElement("option");
  option.value = m;
  option.textContent = m;
  month.appendChild(option);
});

for (let i = 2026; i >= 1950; i--) {
  let option = document.createElement("option");
  option.value = i;
  option.textContent = i;
  year.appendChild(option);
}

function showError(input, message) {
  input.classList.add("input-error");
  input.classList.remove("input-success");
  const el = document.getElementById(input.id + "Error");
  if (el) el.textContent = message;
}

function showSuccess(input) {
  input.classList.remove("input-error");
  input.classList.add("input-success");
  const el = document.getElementById(input.id + "Error");
  if (el) el.textContent = "";
}

function validatePassword() {
  const value = password.value;
  if (!value) { showError(password, "Password is required."); return false; }
  if (value.length < 8) { showError(password, "Password must contain at least 8 characters."); return false; }
  if (!/[A-Z]/.test(value)) { showError(password, "Add at least one uppercase letter."); return false; }
  if (!/[a-z]/.test(value)) { showError(password, "Add at least one lowercase letter."); return false; }
  if (!/[!@#%&*().,?:{}|<>]/.test(value)) { showError(password, "Add at least one special symbol."); return false; }
  showSuccess(password);
  return true;
}

function validateRepassword() {
  if (!repassword.value) { showError(repassword, "Please confirm your password."); return false; }
  if (password.value !== repassword.value) { showError(repassword, "Passwords do not match."); return false; }
  showSuccess(repassword);
  return true;
}

function validateBirthday() {
  const error = document.getElementById("birthdayError");
  if (!day.value || !month.value || !year.value) {
    day.classList.add("input-error");
    month.classList.add("input-error");
    year.classList.add("input-error");
    error.textContent = "Please select your complete birthday.";
    return false;
  }
  day.classList.remove("input-error");
  month.classList.remove("input-error");
  year.classList.remove("input-error");
  error.textContent = "";
  return true;
}

firstname.addEventListener("input", () => {
  firstname.value.trim() ? showSuccess(firstname) : showError(firstname, "First name is required.");
});

```

```

});
lastname.addEventListener("input", () => {
    lastname.value.trim() ? showSuccess(lastname) : showError(lastname, "Last name is required.");
});
username.addEventListener("input", () => {
    if (!username.value.trim()) showError(username, "Username is required.");
    else if (username.value.trim().length < 4) showError(username, "Username must contain at least 4 characters.");
    else showSuccess(username);
});
email.addEventListener("input", () => {
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!email.value.trim()) showError(email, "Email address is required.");
    else if (!pattern.test(email.value.trim())) showError(email, "Enter a valid email address.");
    else showSuccess(email);
});
website.addEventListener("input", () => {
    const pattern = /^(https?:\/\/)?([w-]+\.[w-]+(v.*)?)/;
    if (!website.value.trim()) showError(website, "Website is required.");
    else if (!pattern.test(website.value.trim())) showError(website, "Enter a valid website address.");
    else showSuccess(website);
});
password.addEventListener("input", () => {
    validatePassword();
    if (repassword.value) validateRepassword();
});
repassword.addEventListener("input", validateRepassword);
day.addEventListener("change", validateBirthday);
month.addEventListener("change", validateBirthday);
year.addEventListener("change", validateBirthday);
terms.addEventListener("change", () => {
    const error = document.getElementById("termsError");
    if (!terms.checked) {
        termsBox.classList.add("terms-error");
        error.textContent = "Please agree to the terms and conditions.";
    } else {
        termsBox.classList.remove("terms-error");
        error.textContent = "";
    }
});

form.addEventListener("submit", (e) => {
    e.preventDefault();
    let valid = true;

    if (!firstname.value.trim()) { showError(firstname, "First name is required."); valid = false; }
    else showSuccess(firstname);

    if (!lastname.value.trim()) { showError(lastname, "Last name is required."); valid = false; }
    else showSuccess(lastname);

    if (!validateBirthday()) valid = false;

    if (!username.value.trim()) { showError(username, "Username is required."); valid = false; }
    else if (username.value.trim().length < 4) { showError(username, "Username must contain at least 4 characters."); valid = false; }
    else showSuccess(username);

    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!email.value.trim()) { showError(email, "Email address is required."); valid = false; }
    else if (!emailPattern.test(email.value.trim())) { showError(email, "Enter a valid email address."); valid = false; }
    else showSuccess(email);

    const websitePattern = /^(https?:\/\/)?([w-]+\.[w-]+(v.*)?)/;
    if (!website.value.trim()) { showError(website, "Website is required."); valid = false; }
    else if (!websitePattern.test(website.value.trim())) { showError(website, "Enter a valid website address."); valid = false; }
    else showSuccess(website);

    if (!validatePassword()) valid = false;
    if (!validateRepassword()) valid = false;

    if (!terms.checked) {
        termsBox.classList.add("terms-error");
        document.getElementById("termsError").textContent = "Please agree to the terms and conditions.";
        valid = false;
    }
});

```

```
    } else {
      termsBox.classList.remove("terms-error");
      document.getElementById("termsError").textContent = "";
    }

    if (valid) {
      alert("Account created successfully! 🎉");
      form.reset();
      form.querySelectorAll("input, select").forEach(el => el.classList.remove("input-success"));
    }
  });
</script>
</body>
</html>
```

Output –

127.0.0.1:5500 says
Account created successfully! 🎉

OK

PERSONAL INFORMATION

First Name *

Sejal

Last Name *

Lende

Birthday *

6

July

2010

ACCOUNT DETAILS

Username *

24070521035

E-mail *

sejal.lende.batch2024@sitnagpur.siu.edu.in

Website *

sejallende.www

Password *

Re-password *

☒ I agree to the terms & conditions

Create My Account ↕

Made by Sejal Lende • 24070521035

Conclusion –

The experiment was successfully performed to understand DOM traversal, manipulation, event handling, and form validation using JavaScript. A To-Do List application with Add, Edit, and Delete features was developed, and a registration form was implemented with dynamic validation and error handling.